Synthetic Aperture Radar and Multispectral Satellite Imagery for Methane Emission Assessment of Rice Farms in Central Luzon, Philippines

Jessah Mei Allard[1], Jann Rovic Cueto[2], Allysa Maxine De Asis[2], Arnel Galut[2], Johanne Mae Obras[2]

[1]Caraga State University – Butuan and [2]Rizal Technological University – Boni

Abstract and Description of Proposed Solution

Rice is a staple crop in many regions and countries, especially in Asia, where 90% of the world's rice is grown (Gutaker et al., 2020). Some countries also produce rice for their own consumption, such as the Philippines. The country has about 4.81 million hectares of land for rice production, out of its 30 million hectares of agricultural land (Wehmeyer, 2015). The Department of Agriculture (DA) reported that both irrigated and rainfed rice systems increased in the last 20 years (Philippine Rice Research Institute, 2022). The country has favorable conditions for rice cultivation, such as temperature, humidity, and rainfall (Philippine Atmospheric, Geophysical, and Astronomical Services Administration (PAG-ASA), 2023).

One of the main rice-producing regions in the Philippines is Central Luzon, also known as Region III. It comprises seven provinces: Aurora, Nueva Ecija, Bulacan, Pampanga, Tarlac, Bataan, and Zambales. It is called the 'Rice Capital of the Philippines' because of its capacity to produce large amounts of rice. According to the Department of Science and Technology - Philippine Council for Industry, Energy, and Emerging Technology Research and Development (DOST-PCIEERD)'s RiceLytics, year 2022 has yielded approximately 19.7 million metric tons of rice and 18% of it can be traced back to Central Luzon (see Figure 1). The amount of yield can also be attributed to the fact that Central Luzon holds the largest contiguous plain in the country (National Nutrition Council - Region III, 2020). The researchers chose this region to measure methane emissions from rice fields, because it contributes significantly to the country's rice supply. Jiahui (2020) found that agriculture accounted for 74.8% of the Philippines' methane emissions (see Figure 2), which are mostly related to rice cultivation.

Provided the threats raised by methane as it contributes to climate change, and the gap in detecting weaker methane emission, the researchers propose a novel approach to the said gap by quantifying methane (CH4) emissions in rice paddy fields in the Philippines through an integrated approach combining remote sensing approaches for land cover mapping and methane detection, ground measurements, and machine learning to both estimate and predict the CH4 emissions.

Due to various factors affecting methane emission detection, the study will have a specified set of parameters to identify. This identified set of parameters will be limited to detecting methane emissions in rice cultivation seasons where weather, climate, or wind influences are not considered. Given that the Philippines is a tropical country affected by multiple typhoons in a year, it is important to note that these will have effects on the methane emission and detection; however, the project will isolate the effects of the identified parameters and obtain more reliable results.

Figure 1. Rice Production in the Philippines as retrieved from Ricelytics (2023) via DOST-PCIEERD.

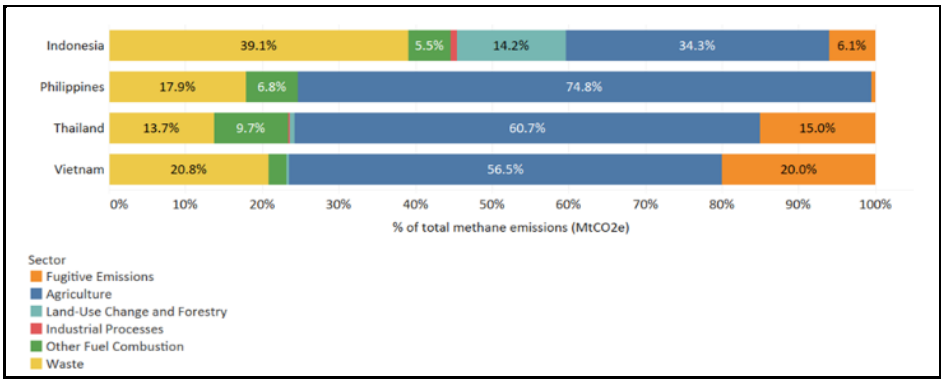

Figure 2. Methane Emission Sources from Southeast Asian Countries as retrieved from Jiahui (2022) via Climate Watch.

General Technical Approach

Satellite Data Specifications

The project will utilize various satellites to collect and gather different data for the parameters. These satellites are Sentinel 1 SAR, Sentinel 2 MSI, Sentinel 5P TROPOMI, and MODIS. Given that the project's objective is to detect weaker sources of methane emissions, the researchers will use an integrated approach to provide a model visually presenting the methane emission sources and the predictions about the affected areas and demographics. To attain this goal, the satellites above will be utilized. First, Sentinel 1 SAR and Sentinel 2 MSI will be used for mapping Central Luzon and identifying the location of rice paddies in the area. Sentinel 1 and Sentinel 2 are used by Inoue, et al. (2020) for the same objective; combining both satellites makes land cover mapping more accurate and effective. Optical images will be gathered using a multispectral imager, and synthetic aperture radar (SAR) will be used to resolve its limitations caused by cloud cover, weather changes, and night-time imaging. It will also supplement data for soil moisture index.

Sentinel 5P TROPOMI will be utilized in the project due to its ability to monitor the atmosphere. This satellite can measure air quality, ozone, methane, UV radiation, climate monitoring, and forecasting in high spatiotemporal resolution. For this mission, Sentinel 5P will monitor the methane emissions using XCH4 or methane flux. Lastly, Terra and Aqua Moderate Resolution Imaging Spectroradiometer (MODIS) will be used to gather data parameters in the form of land surface

temperature, normalized difference vegetation index, land surface emissivity, potential evapotranspiration, and evapotranspiration.

Given the need to utilize various satellites, the researchers of this project have a specified set of information regarding the parameters that will be gathered and used in the project. Each satellite, Sentinel 1, Sentinel 2, Sentinel 5P, and MODIS, will collect different datasets, which will be analyzed to make a model and predict the methane emissions in Region III, Central Luzon. The following satellite specifications, resolution, and frequency are shown in Table 1.

Satellite	**Specifications**		**Resolution**	**Satellite Revisit Time**	**Frequency of Data Gathering**
	Band	**Usage**			
Sentinel 1 – Synthetic Aperture Radar (SAR)	C	Interferometric Wide Mode for Land Cover Mapping and Soil Moisture Index	10 m	5 to 6 days	Mean average every 14 days
Sentinel 2 – Multispectral Imager (MSI)	B2, B4, B8, and B 11	Blue, Red, Near-Infrared, Short-wave Infrared 1 for Land Cover Mapping	10 m (B2, B4, and B8) 20 m (B11)	5 to 6 days	Mean average every 14 days
Sentinel 5P – Tropospheric Monitoring Instrument (TROPOMI)	UV1, Near Nadir and SWIR	Ultraviolet 1 and Shortwave Infrared for Methane Flux (XCH4)	7 x 3.5 km (All spectral bands) 7 x 28 km (UV1) 7 x 7 (SWIR)	1 to 2 days	Mean average every 14 days
Terra and Aqua Moderate Resolution Imaging Spectroradiometer (MODIS)	1, 2, 3, 4, 6, 7, 31, 32, 33, 34, 35, 36	Shortwave SWIR, VIS, and NIR, Longwave Thermal Infrared TIR for Air Quality Detecting Methane	250 m (1 and 2) 500 m (3, 4, 5, 6, and 7) 1000 m (31, 32, 33, 34, 35, and 36)	1 to 2 days	Mean average every 14 days, spanning from 2018 to 2023

Table 1. Satellite Data Specification, Resolution, and Frequency.

Ground Measurement Data Specifications

The project will utilize a low-cost version of the method conducted in the study of Karki, et al. (2021) to measure greenhouse gas fluxes with static flux chambers. This low-cost static gas chamber comprises a PVC foundation, adjustable extensions (15.3-91.4%), and a chamber lid. A fan, vent tube,

and thermocouple wire are installed on the lid to measure the air temperature. As for the PVC, four holes of 11 cm in diameter are drilled into the bottom of each 29.5 cm x 22.9 cm. This will help with the water flow and root contact. This chamber also facilitates water circulation inside the collars, which will be conducted by inflicting two holes (2.86 cm) in the upper portions. Rubber stoppers are used to seal the holes for gas sampling. Once the rice has three to four leaves, it is estimated to have a set firm ground at 15 cm in depth. Boardwalks are implemented for each collar to reduce soil and plant disturbance during sampling.

For this project, the ground-based data (g/m²/day or kg/ha/season) will be gathered from this instrument (as adapted from the methods of Muhammad Faseeh Iqbal et al., 2023), measuring CH4 emissions twice a week for 1-2 months. Table 2 shows the specifications of the low-cost static gas chamber.

Figure 3. Low-cost Static Gas Chamber from 'Guidelines for Measuring CH4 and N2O Emissions from Rice Paddies by a Manually Operated Closed Chamber Method,' (n. d.).

Material	Specification	Usage
Microcontroller	ESP32	This device is a 32-bit LX6 microprocessor with one or two cores and a maximum frequency of 240 MHz. It has 520 KB of SRAM, 448 KB of ROM and 16 KB of RTC SRAM. It supports Wi-Fi 802.11 b/g/n, Classic Bluetooth v4.2 and BLE. It has 34 GPIOs, 18 channels of ADC, 2 channels of DAC and various serial interfaces. It also features Ethernet MAC, SD/SDIO/MMC host and slave, motor and LED PWM, secure boot, flash encryption and cryptographic hardware acceleration.
Gas Sensor Module	MQ4	These devices can detect CH4, natural gas, and LNG leaks in residential and industrial settings, and are not affected by alcohol vapors, cooking odors, or tobacco smoke. They are commonly used in gas leak detection systems.
Humidity and Temperature Sensor	DHT11	This has a calibrated digital output of temperature and humidity with its sensor system. It also has high accuracy and long-term stability due to its unique digital signal processing, includes a resistive humidity sensor, an NTC temperature sensor, and a high-performance 8-bit microcontroller.

Table 2. Specifications of the Low-cost Static Gas Chamber.

Integrated Approach: Satellite and Ground Measurement Methane Detection

The project aims to estimate and predict methane emissions in rice cultivation using a machine learning algorithm trained with ground data. Historical data will also be utilized upon its availability. This approach was adapted from Akumu, et al. (2010), and Sun, et al. (2017), which specifically made models for rice paddies and wetlands. With the integration of remote sensing approach and ground measurement approach, the project has an improved data gathering and validation, as well as model accuracy. In height of its significance, the project will provide a CH4 assessment model and prediction in Region III, Central Luzon, Philippines, which will help supplement evidence-based policies and sustainable practices for mitigation and addressing its impacts on climate change.

Both satellite measurements and ground-based data measurements have a specified frequency of data gathering. This varying set of frequency are based on data availability, as well as in situ data gathering. Moreover, in order to enhance the dataset's reliability and address any data limitations, data engineering techniques and data imputation methods will be also employed. The main motivation is to use the augmented data as a regularized to reduce the potential over-fitting of the trained models (Viera Maslej-Krešňáková et al., 2022). These processes involve modifying and expanding the dataset to provide a more robust foundation for training and evaluating the machine learning model. By incorporating this comprehensive dataset, the model will be equipped to make accurate predictions of CH4 emissions during the growing season in rice cultivation.

As for model building and refinement, the project researchers will rely on Python, a versatile programming language. In machine learning, it's standard practice to divide data into two sets: one for training and the other for testing (see Table 3). This division helps us assess how well the model can work with new, unseen data, ensuring its performance in real-life scenarios. Importantly, once the model is trained, we plan to apply it to historical data (e. g. 2019) annually. This will help us spot and minimize any differences between the model's predictions and actual outcomes. By applying these established practices and drawing insights from reputable sources, the project aims to contribute to the field of agricultural monitoring and land use analysis, providing evidence-based findings that can benefit decision-makers and practitioners in the agricultural sector.

As for the prediction of the methane emissions, the project will set its foundations on the available historical data with at least 5 years average, in all the specified parameters that are presented in Table 3. This will help suggest and improve the model prediction, which has an improvement potential in the years to come. It is also important to note the necessity of the prediction, given that there are affected demographics surrounding the rice paddies and cultivation areas. Table 4 shows the demographics in Central Luzon, which highlights seven provinces (two provinces are highly urbanized). From this data, the population density for Central Luzon is at 567 inhabitants per km^2 or 1,469 inhabitants per mi^2. In addition, for this prediction, the project will validate its predicted results by reducing the error through data testing. Using the parameters, the researchers will test the data from the previous years and validate the prediction, until the error is reduced, and the prediction is accurate.

Data Parameter	Definition	Formula	Citation
Land Surface Temperature (LST)	This is a key factor positively correlating with increased CH4 emissions in rice cultivation, given that higher temperatures pave the way for microbial activity and methane generation to exist in paddy fields. CH4	LST = DN *0.02 – 273.15 $Ts = \frac{K_2}{(ln\frac{K_2}{L_\lambda}+1)}$	Wan (2013)

	emissions tend to be higher due to microbial decomposition of organic matter in waterlogged rice paddies.		
Normalized Difference Vegetation Index (NDVI)	THE NDVI is selected to be able to monitor the vegetation to balance the changing illuminations, aspect, surface slope, and other extension factors. By using NDVI for remote sensing, the model will gain insights into how vegetation health affects the methane production in rice paddies.	NDVI = ((IR - R) / (IR + R))	Malik (2014) and Yadav & Borana (2019)
Land Surface Emissivity (LSE)	This will be extracted from the land temperature and vegetation index that will be used to measure empirical relationship between NDVI and LSE equation from emission and spectral reflectance.	$\varepsilon = 1.0094 + 0.047 \ln (NDVI)$	(United States Geological Survey, 2019)
Evapotranspiration	Evapotranspiration influences water vapor releases causing increase in CH4 anaerobic cycles which allow the model to account for the water vapor in methane emission dynamics.		Shukla et al. (2021)
Atmospheric CH4 Mixing Ratio (ppb)	Data that will indicate the concentration of CH4 in the atmosphere in rice paddies. These data are needed to be able to gain further understanding of the overall contribution of atmospheric methane in the study area.		Kim et al. (2020)
Soil Moisture (%)	Accurate assessment of soil moisture, obtained from remote sensing sources helps the model account for local variations in waterlogged conditions such as rice fields which significantly influence methane emissions.		Farahani et al. (2022)

Table 3. Remote Sensing Data Parameters.

Name	Cities	Municipalities	Barangays	Area Approximate	Population (2020)	Density (2020)
Aurora	0	8	151	3,133.40 km2 (1,209.81 sq mi)	235,750	75 / km2 (195 / sq mi)
Bataan	1	11	237	1,372.98 km2 (530.11 sq mi)	853,373	622 / km2 (1,610 / sq mi)
Bulacan	3	21	569	2,783.69 km2 (1,074.79 sq mi)	3,708,890	1,332 / km2 (3,451 / sq mi)
Nueva Ecija	5	27	849	5,689.69 km2 (2,196.80 sq mi)	2,310,134	406 / km2 (1,052 / sq mi)
Pampanga	2	19	505	2,001.22 km2 (772.68 sq mi)	2,437,709	1,218 / km2 (3,155 / sq mi)

Tarlac	1	17	511	3,046.49 km2 (1,176.26 sq mi)	1,503,456	494 / km2 (1,278 / sq mi)
Zambales	1	13	230	3,630.35 km2 (1,401.69 sq mi)	649,615	179 / km2 (463 / sq mi)

Table 4. Central Luzon Demographics.

Conceptual Framework

The project will utilize remote sensing satellite data and ground measurement data as parameters for machine learning. Once these methods are conducted, the projected output of this project is a CH4 Estimation Model and Prediction. The subsequent figures presented at this part of the paper is the Conceptual Framework. This will highlight the entire process of the project proposal.

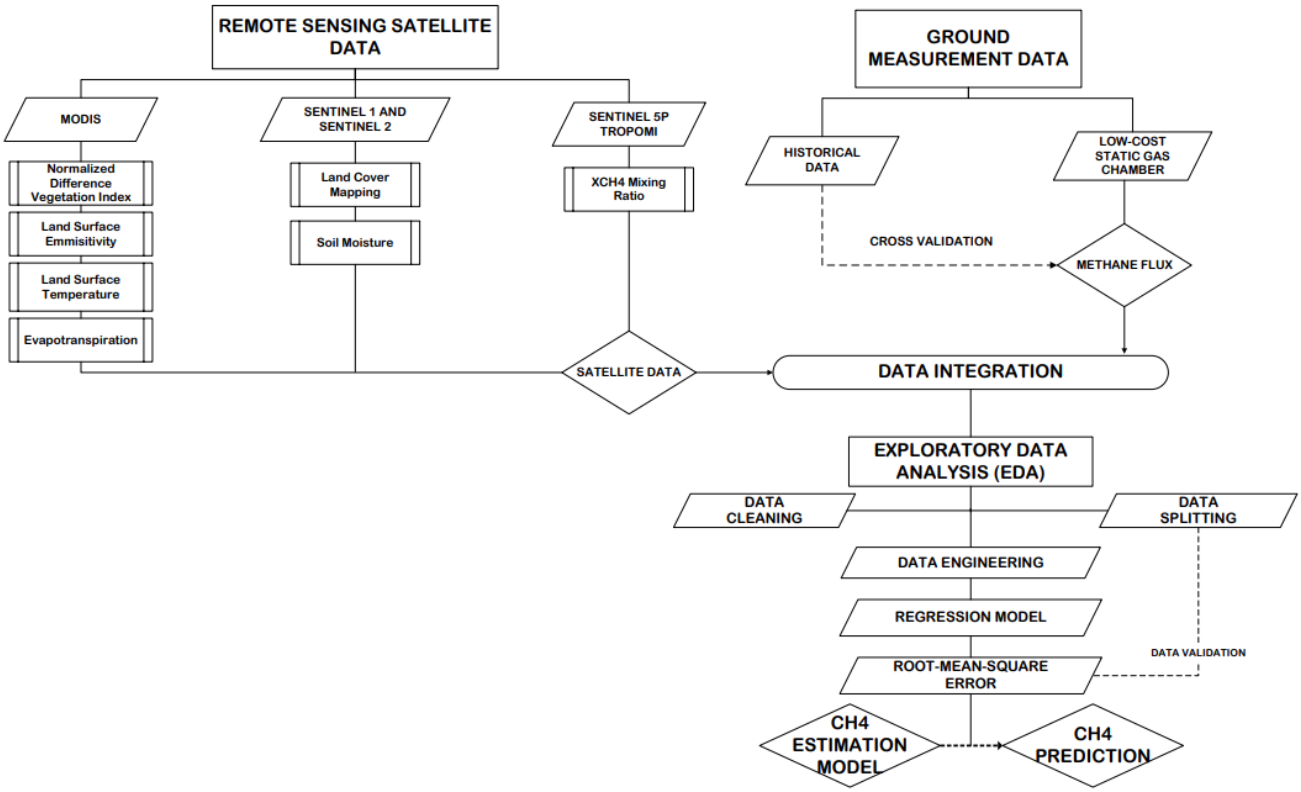

Figure 4. Conceptual Framework.

Request for Support

The project will be conducted to detect weaker methane emissions in Central Luzon, Philippines. This will be done with a combination of remote sensing, ground measurements, and machine learning. To carry out the project successfully, they need certain support and requests from the SpaceBase and related communities and providers, as shown in Table 5.

Support Needed	Description	Anticipated Provider

ArcGIS Professional License	An advanced software which will be used by the researchers for land cover mapping and machine learning.	SpaceBase and/or partner communities
Historical Data and Ground Measurement of XCH4 Flux Data during the Rice Cultivation Season	The researchers will need the statistics of the historical data and ground measurement during the rice cultivation season, in order to specifically input the numbers for machine learning.	Department of Science and Technology (DOST), Philippine Rice Institute (PRI), and Department of Agriculture (DA) as supported and endorsed by SpaceBase and/or Philippine Space Agency (PhilSA)
Knowledge or Training Support on Implementing, Extracting, and Quantifying CH4 from the Static Gas Chamber	The researchers will be utilizing a static gas chamber to detect methane emissions in-situ, and with this, they will be needing support on how to properly implement the instrument, extract the data, and quantify in order to validate and use correlational analysis with the remotely sensed data.	Department of Science and Technology (DOST), Philippine Rice Institute (PRI), and Department of Agriculture (DA) as supported and endorsed by SpaceBase and/or Philippine Space Agency (PhilSA)
Static Gas Chamber Analyzer	This will be an instrument proposed by the researchers to detect methane emissions in Central Luzon. This will be placed on the rice paddies with and upon ethical considerations and agreements with concerned locals. The researchers will need the help of agencies who already owns such instrument.	Department of Science and Technology (DOST), Philippine Rice Institute (PRI), and Department of Agriculture (DA) as supported and endorsed by SpaceBase and/or Philippine Space Agency (PhilSA)
List of Rice Farmers or Rice Paddy Locations in Central Luzon	The researchers will need the help of the local government units of Central Luzon by gathering the list of rice farmers in the region. This will help the researchers specify and validate the remotely sensed data.	Central Luzon Local Government Units (LGUs) as supported and endorsed by the Department of Agriculture, SpaceBase, and/or Philippine Space Agency (PhilSA)
Financial Support for Transportation, In-situ Oculars, and Experimentations	The researchers are in the National Capital Region (four researchers) and Mindanao (one researcher) and are geologically further from the specified location for the study. In this regard, the researchers will need financial support (or any other form of support) which will be utilized for transportation, in-situ oculars, and experimentations provided with a liquidation report.	SpaceBase and/or Philippine Space Agency (PhilSA)

Table 5. Support and Requests with the Support, Description, and Anticipated Provider for the study.

Projected Plan and Schedule

The researchers aim to develop innovative solutions for detecting weaker methane emissions in Central Luzon, Philippines. The challenge consists of two phases: Phase 1 and Phase 2.

In Phase 1, the study will have a research design and planning process, where they will identify the problem, the objectives, the methods, and the expected outcomes of their solution. Then, the researchers will also conduct a literature review to survey the existing knowledge and approaches on the topic. The output of Phase 1 will be a written report that summarizes the solution proposed for the challenge.

In Phase 2, the participants will implement their solution by collecting and analyzing data from various sources, such as satellites. These will be processed using reliable software such as ArcGIS, QGIS, Python and SPSS. Afterward, the researchers will also demonstrate the solution and how it will be able to detect weaker methane emissions effectively and efficiently. The output of Phase 2 will present the process for data collection and analysis, as well as the practical demonstration of the solution. The projected plan and schedule are presented in Table 6.

PHASE 1					
Stage	**Activity**	**Estimated Duration**	**Start Date**	**End Date**	**Deliverables**
Research Design and Planning	Finalize target areas of methane emission and satellite data to be use	1 week	July 1	July 8	Confirmed the target areas and satellite data to be use
	Develop research design to detect and monitor methane emission	1 week	July 9	July 16	Draft framework for method and data parameters
	Prepare research proposal	2 months	July 1	August 30	Research proposal and consultation notes with experts
Literature Review	Search and synthesize relevant literature	1 month	July 1	August 1	Notes and written output from the review process
	Prepare preliminary review of related literature				Draft of literature review to be added in research proposal
	Identify the research gaps that will be address in the project				Include the gaps that need to be address
Write Up	Submit proposal to Space Base Challenge 2023	1 day	Aug 31, 2023		Final submission of proposal
PHASE 2					
Data Collection	Finalize the data collection plan	1 month	September 4	October 8	Sample data collection plan
	Pre-test the pilot data collection				Finalized data collection instrument
	Carry out the data collection				Raw data

	Write up data collection				Draft data collection section of the paper
Data Analysis	Prepare data for analysis	2 – 3 months	October 9	December 30	Data ready for analysis
	Analyzed remote sensing +GIS data for target area selection modelling from Sentinel-1 and Sentinel-5				Spatial data of CH3 mixing ratio, soil moisture, and land emissivity
	Using machine learning, evaluate the CH4 estimation in target areas				Statistical output on correlation between CH4 emission in rice farms
	Draw correlational analysis between ground measurement and satellite data				Estimation of CH4 from remote sensing and ground measurement
Practical Demonstration of Solution	Co-designed with local researchers, government, and farm communities for compatibility	1 month	January 1	January 31	Address the necessary mitigation and actions to reduce methane emission in the field
	Support reduction of methane emission projects				Continuous support to local government
Write Up	Final draft of paper and editing	2 months	December	January	Final draft
	Review draft and revision	1 month	January	February	Notes of feedback
	Submission	1 day	February 25, 2024		Submission of application and paper

Table 6. Projected Plan and Schedule of the Study.